4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

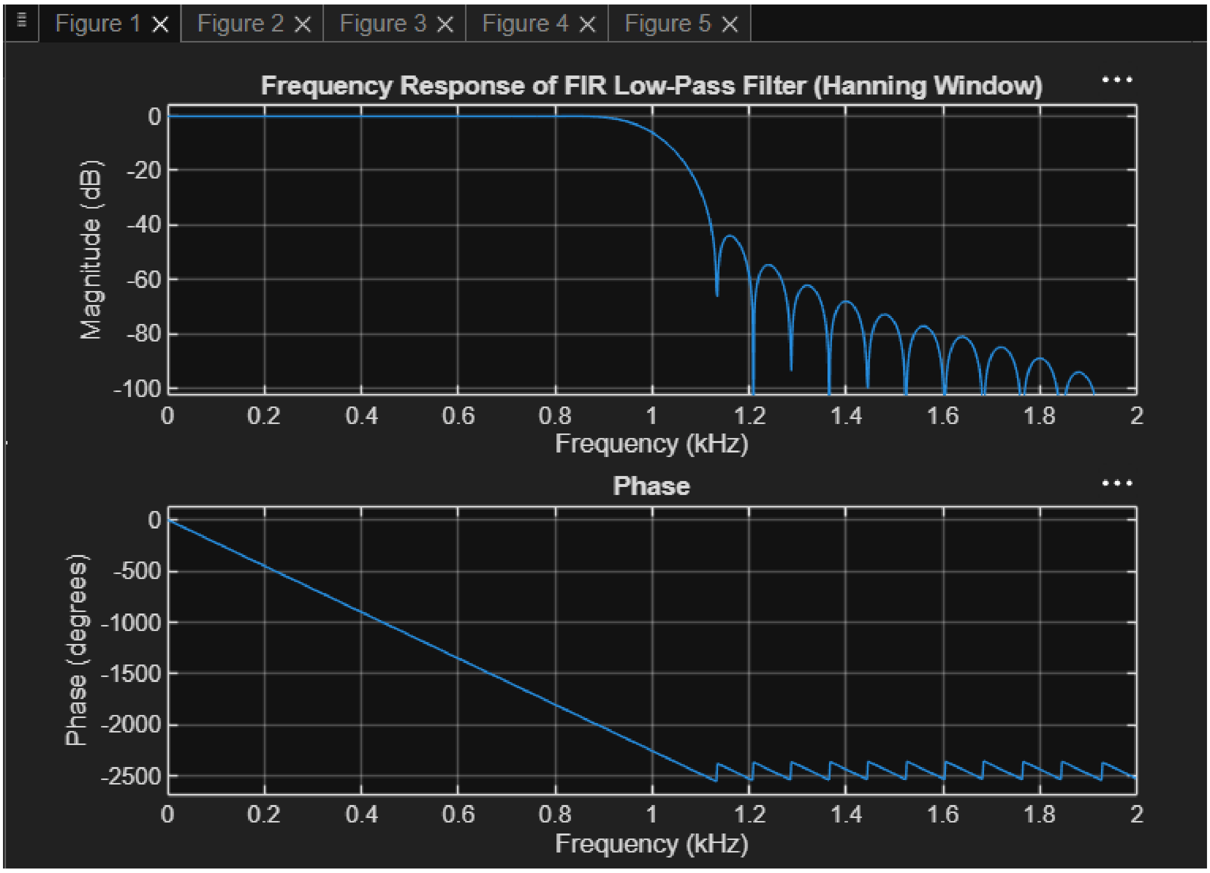
CODE:

```
clc;
clear;
close all;

%% Input Parameters
fp = 1000;        % Passband cutoff frequency (Hz)
Fs = 4000;        % Sampling frequency (Hz)
N = 50;           % Filter order

% Normalized cutoff frequency
Wn = fp/(Fs/2);

%% Design FIR Low-Pass Filter using Hanning (Hann) Window
b = fir1(N, Wn, 'low', hann(N+1));

%% Display Filter Coefficients
disp('FIR Low-Pass Filter Coefficients (Hanning Window):');
disp(b');

%% Frequency Response
figure;
freqz(b,1,1024,Fs);
title('Frequency Response of FIR Low-Pass Filter (Hanning Window)');
```

```
%% Magnitude Response
[H,f] = freqz(b,1,1024,Fs);

figure;
plot(f,20*log10(abs(H)),'LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response (Hanning Window)');
axis([0 Fs/2 -100 5]);

%% Phase Response
figure;
plot(f,unwrap(angle(H)),'LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (radians)');
title('Phase Response (Hanning Window)');

%% Impulse Response
figure;
impz(b,1);
grid on;
title('Impulse Response (Hanning Window)');
```

```
%% Pole-Zero Plot

figure;

zplane(b,1);

grid on;

title('Pole-Zero Plot (Hanning Window)');


%% Display Filter Information

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %d Hz\n',fp);

fprintf('Sampling Frequency = %d Hz\n',Fs);
```

OUTPUT:



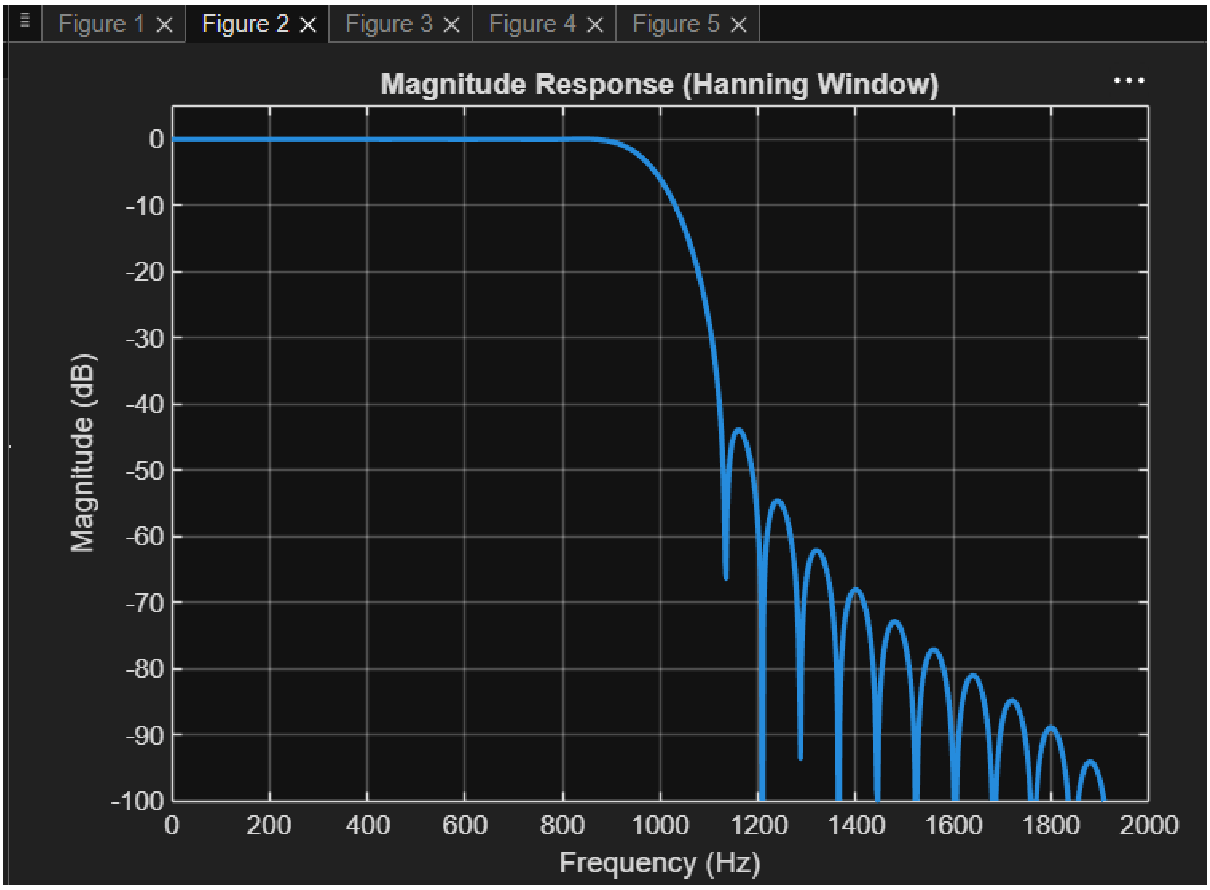
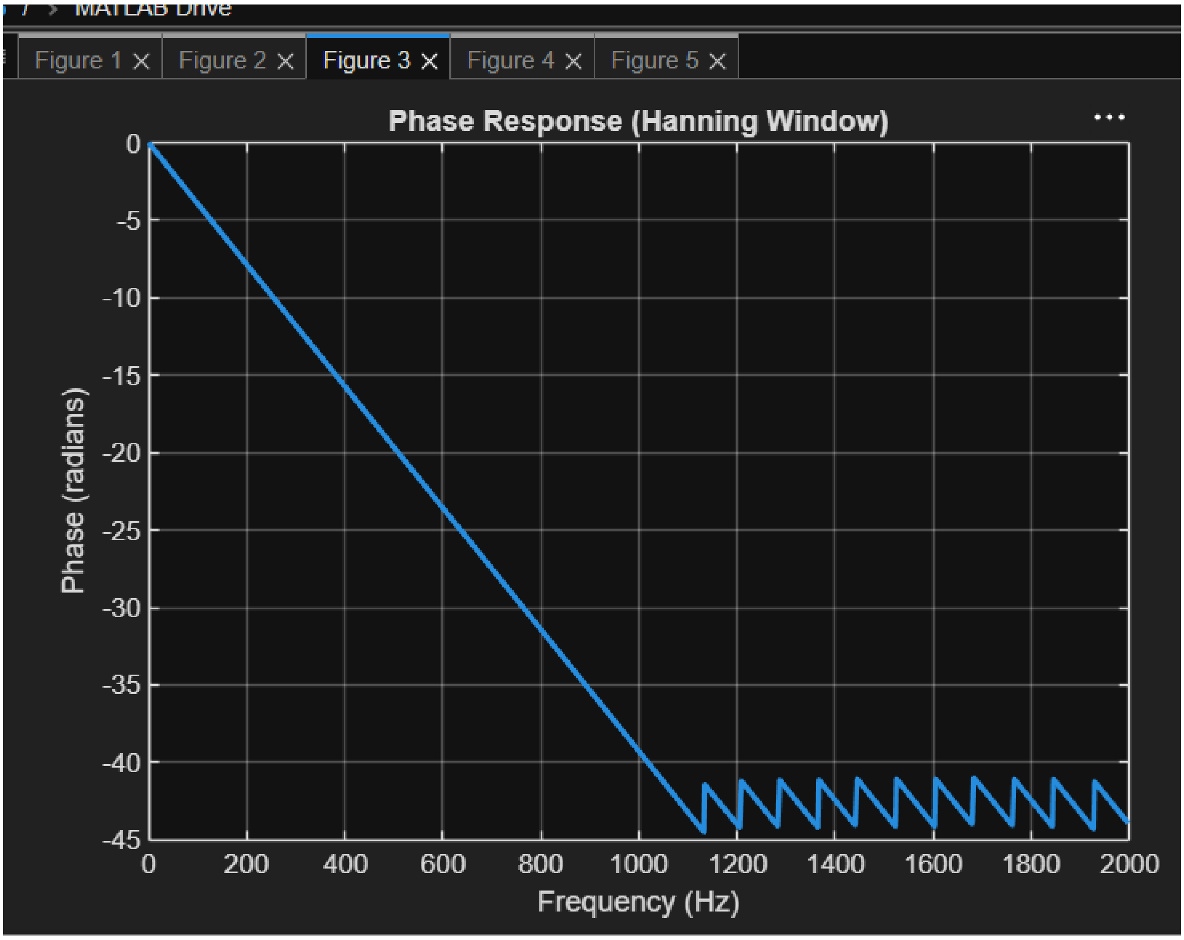
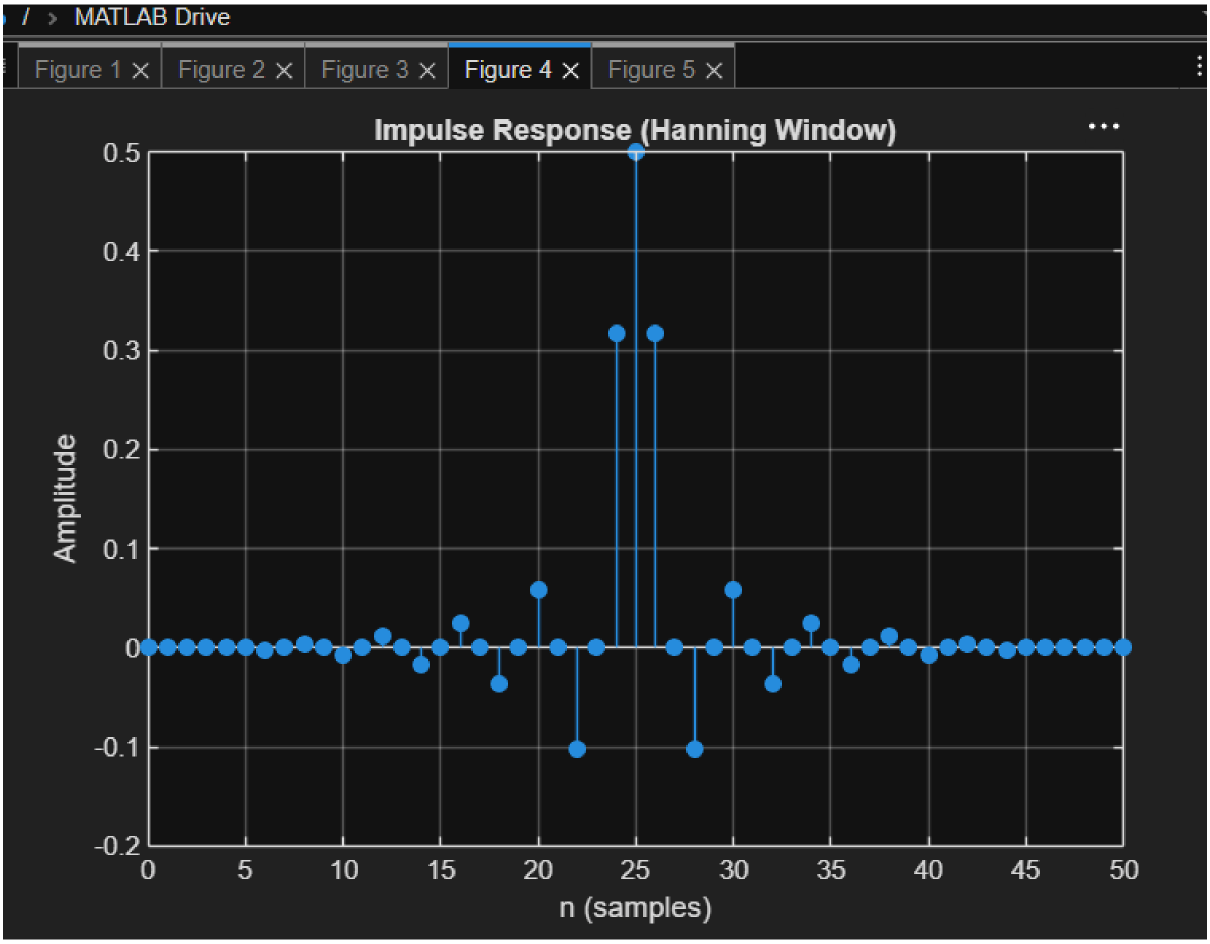


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Magnitude Response (Hanning Window)
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Frequency (Hz)



MATLAB Drive
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Phase Response (Hanning Window)
Phase (radians)
Frequency (Hz)
0
-5
-10
-15
-20
-25
-30
-35
-40
-45
0
200
400
600
800
1000
1200
1400
1600
1800
2000


MATLAB Drive
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Impulse Response (Hanning Window)
Amplitude
n (samples)
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
5
10
15
20
25
30
35
40
45
50

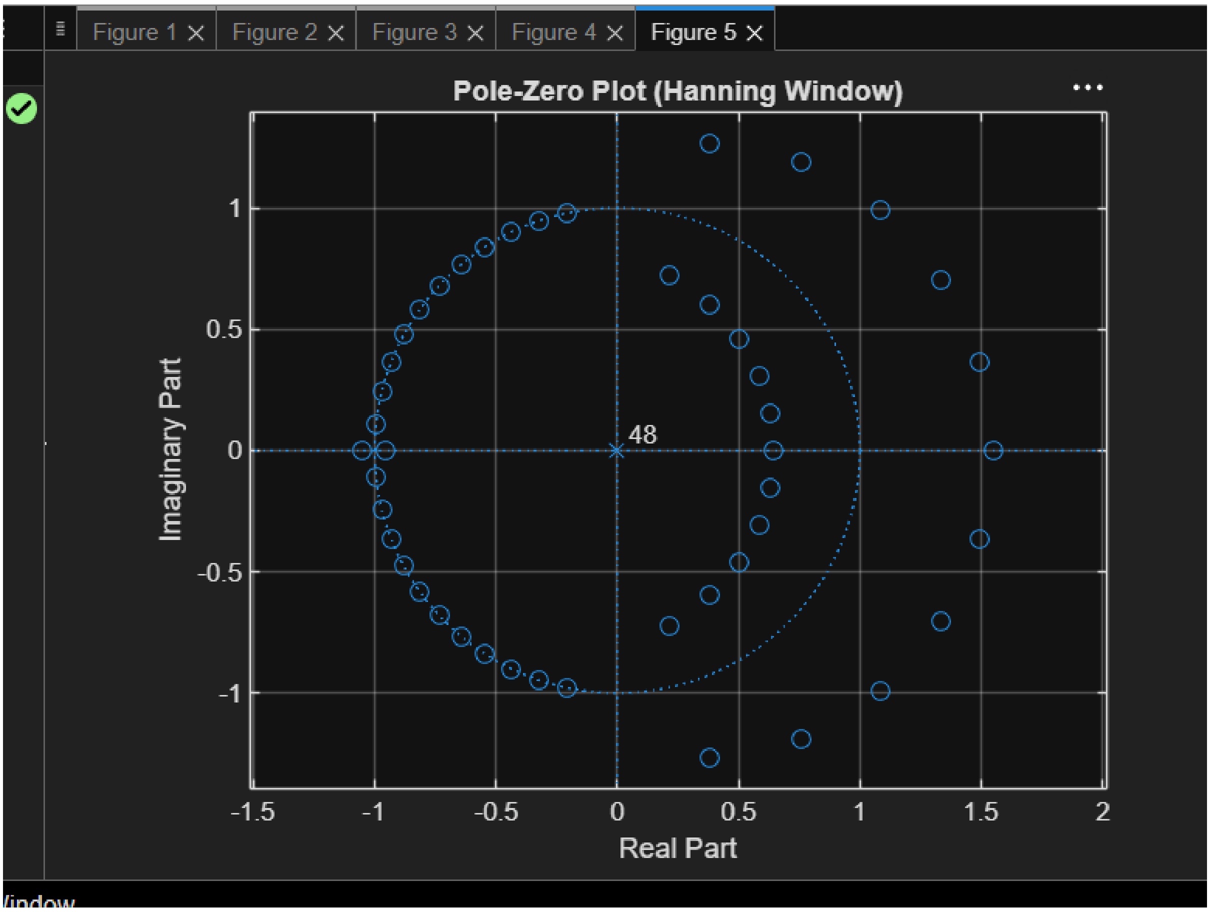


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Pole-Zero Plot (Hanning Window)
Imaginary Part
Real Part
48
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1
1.5
2